Google Cloud H2S

AI Builder Cup

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Team Details

Team name: SKUdesk

Team leader name: Mr. Chonlarnatt Poonchattakarn

Team member: Ms. Ploypitcha Boontavipitak

Problem statement: Shoppers and small resellers cannot tell whether a cheaper listing is the identical product, so a “deal” is often a different model, a case or a clone. SKUdesk finds the cheapest landed cost, prices a resale and checks arbitrage only on provably identical products.

Brief about the idea

- **One agent, three jobs.** It finds the cheapest landed cost, prices a resale and checks arbitrage, but only when two listings are the identical product.
- **Gemini is the identity gate.** It must quote the brand and model it relied on, and our code checks that the quotes really appear in the listing. A second Gemini step looks at the photo and can only take a match away.
- **The honest “no” is the product.** No identical product, or no gap left after fees, returns and postage: the agent says so and shows every listing it turned down. A wrong “yes” loses money.
- **A person approves every spend.** The agent never buys anything itself. Each buy is a proposal; an approved resale can then be listed in Google Merchant Center by the owner.
- **Real services, live now.** Google Shopping results, Gemini, Cloud Run, Firestore, Cloud Scheduler, Secret Manager and the Merchant API, at skudesk.lol.

Opportunities

- **How different is it?** Price trackers compare titles and show a “lowest price” that is often another model, an accessory or a clone. SKUdesk makes identity the first gate and shows every listing it turned down and why.
- **How does it solve the problem?** A two-step identity check (text with quoted evidence, then the photo), a price-sanity filter for impossible prices, and a cost model in integer cents: landed cost, 13% marketplace fee, 2% returns reserve, outbound postage and a margin floor.
- **USP** The honest “no”. 67 labelled product pairs ran against the real model with zero wrong matches, and a person approves every spend. A market view shows asks, four buyer policies’ bids and the spread, so the reason for a “no” is visible.

Measured on the live service (18 Oct 2026, 12 products watched)

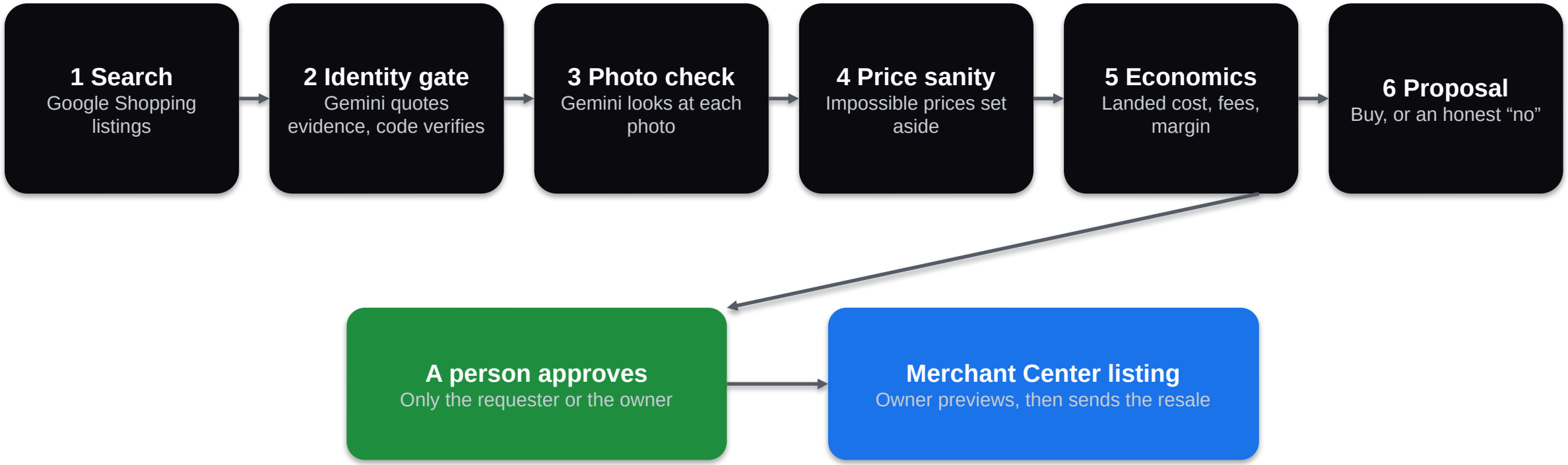
249	201	14	$6,995
listings screened by the identity gate	turned down: not the same product (81%)	impossible prices set aside as bad data	of fake discounts the agent refused to trade on

Figures come from the run-metrics page at skudesk.lol and keep growing as the scheduler checks the watchlist.

List of features offered by the solution

- **Identity gate** Gemini quotes brand and model; code verifies the quotes and reads storage size and model tier itself.
- **Photo check** Gemini compares each matching photo with the requested product and can only take a match away.
- **Price sanity** Prices far below the median or recalled retail are set aside as bad data.
- **Economics** Landed cost, fees, returns, postage and margin floor, in integer cents.
- **Approval** Every buy is a proposal; only the requester or the owner decides.
- **Market page** Order book of identical asks, four buyer-policy bids, spread, history chart, 3D product view.
- **Watchlist and scheduler** Up to 12 products checked on a schedule, longest-waiting first.
- **Live agent view** A graph that lights up as the agent searches, judges, looks and prices.
- **Resale listings** Preview and send an approved resale to Google Merchant Center (owner only).
- **Owner controls** Per-buy limit, margin floor, kill switch, audit log.

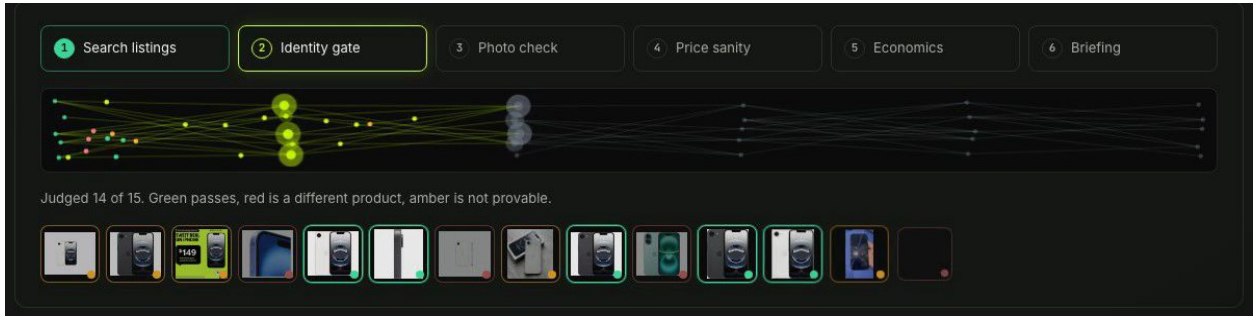
Process flow diagram



Nothing identical, or no gap after costs: the agent says no and shows what it turned down. The scheduler runs the same flow on the watchlist.

The agent at work: one real check

LEGO Technic Ferrari Daytona SP3 (42143), checked on skudesk.lol with real Google Shopping listings and real Gemini verdicts.



Live agent graph, from the demo recording (a check of an iPhone 16e). Every dot is one listing moving through six stages: green = identical, red = a different product, amber = not provable.

40	30	6
listings found	judged by Gemini	identical
20	**4**	**3**
different product	not provable	bad prices set aside

Of the 6 identical, 3 had prices that cannot be real ($5.83, $29.00 and $200.00 for a set that retails near $449.99), so only 3 were priced.

Five listings the agent did not trade on, and why

Listing	Verdict	Why
Ferrari Daytona SP3 “The Sense of Perfection” book	Different	A book, not the set
Brickshell Cases Wall Display Frame	Different	An accessory, not the set
“KING 8888 compatible 42143” building blocks	Different	Other brand: a clone
eBay 42143 “Collectible Supercar Building Kit”	Not provable	Brand not found in the listing
eBay LEGO Ferrari 42143 at $5.83	Set aside	Impossible next to $449.99 retail

Buy **BrickFever $525.00 landed** ($515.00 + $10.00 assumed shipping)

Resell at **$700.42** (next cheapest identical ask)

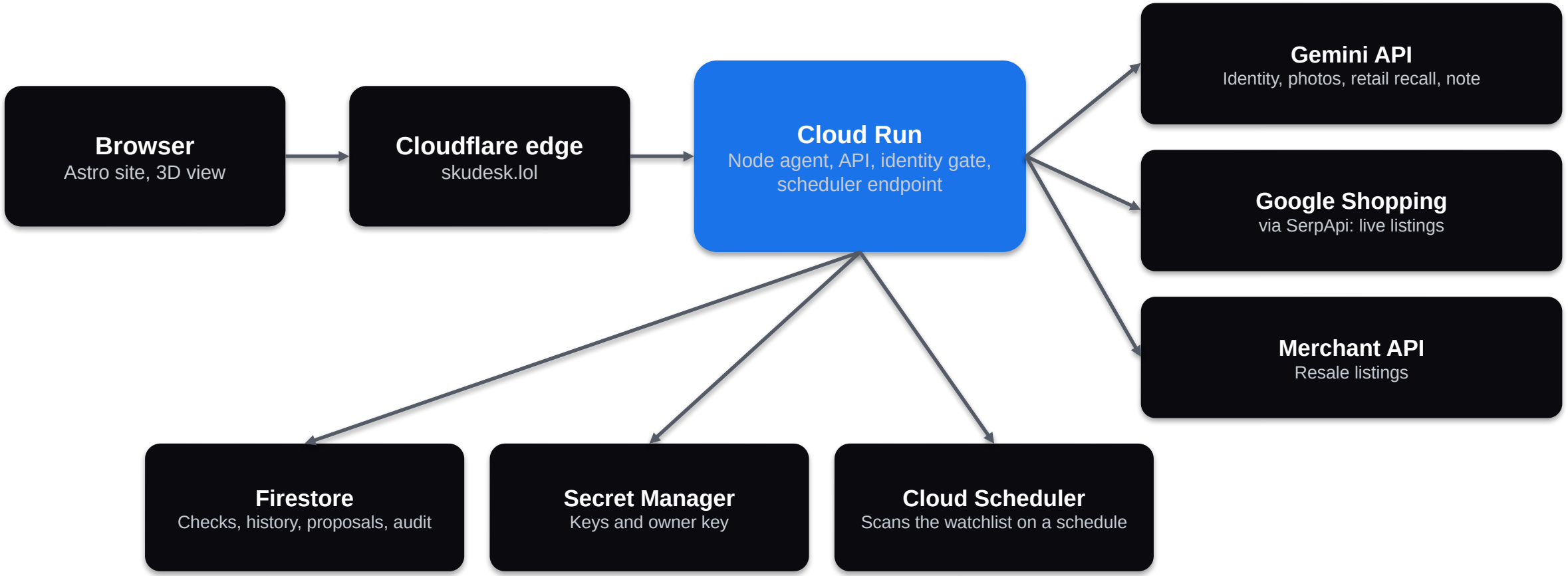
Costs **13% fee $91.05, 2% returns $14.01, postage $15.00**

Net **+$55.36 (10.5%)**, above the 10% margin floor

A proposal now waits for a person. The agent does not buy.

Asking prices, not sold prices. Checked 16 Oct 2026.

Architecture diagram of the proposed solution

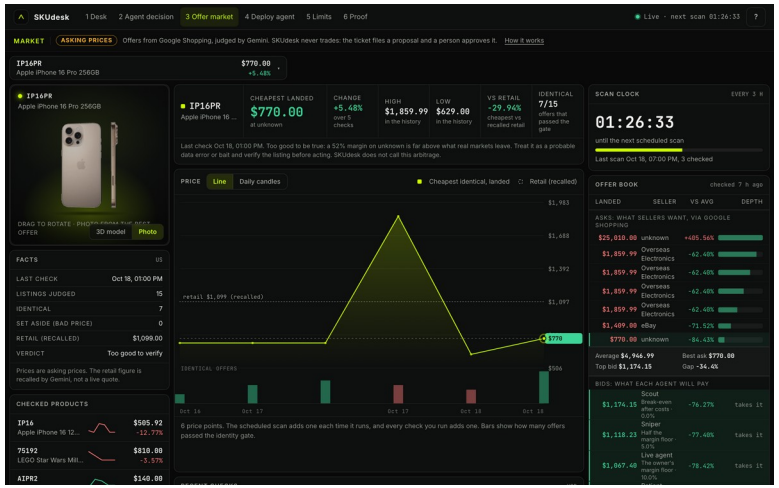


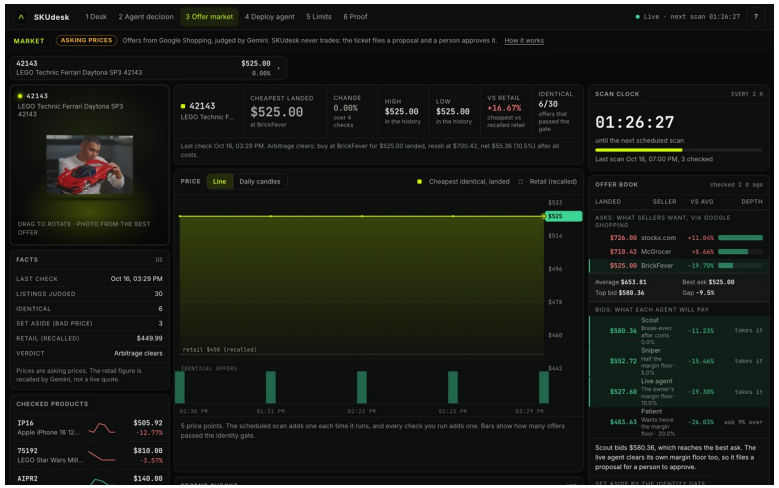
The edge worker adds a shared secret so the service only answers traffic that came through it. The page content policy allows scripts from the site only.

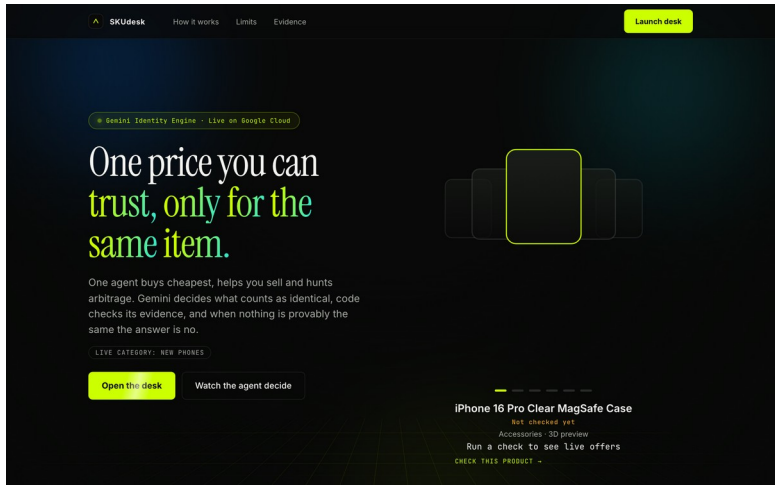
Technologies used in the solution

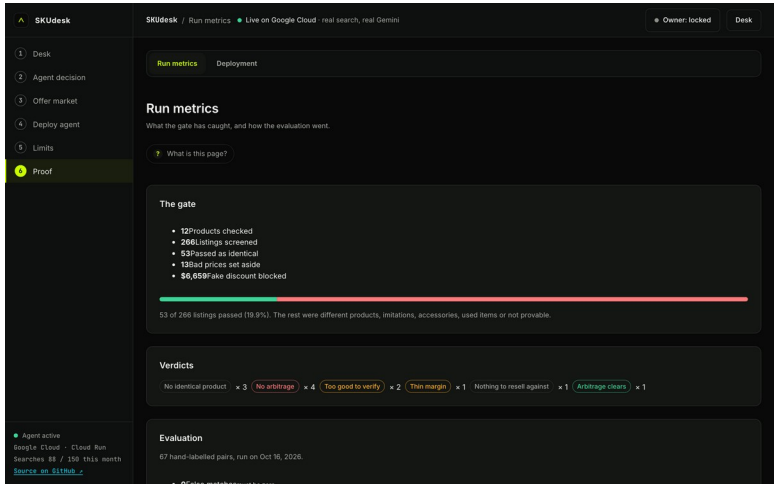
- **Google Cloud** Cloud Run, Firestore, Cloud Scheduler, Secret Manager, Merchant API
- **Gemini** Google Gen AI SDK: structured identity verdicts, image understanding, grounded retail recall
- **Google Shopping** Live listings through SerpApi
- **Backend** Node.js with native TypeScript, no framework, node:test
- **Frontend** Astro, vanilla TypeScript, WebGL 3D view, canvas
- **Quality** 136 unit tests, 21 browser end-to-end tests (Playwright), a labelled evaluation set, a smoke test for the deployed service

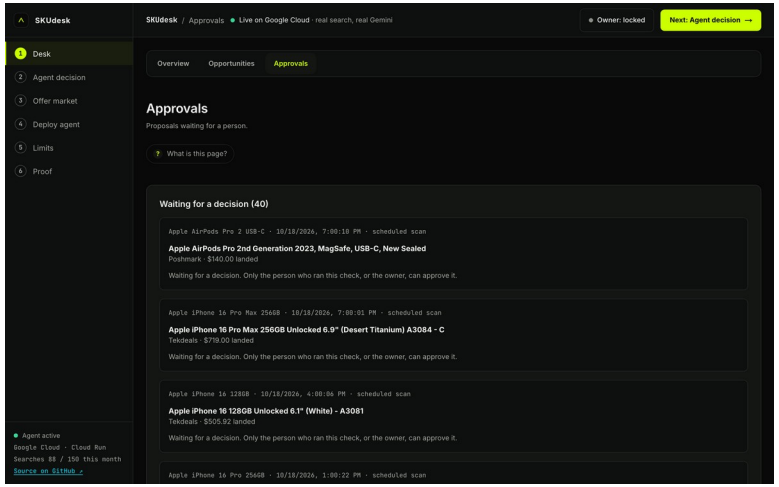
Snapshots of the prototype

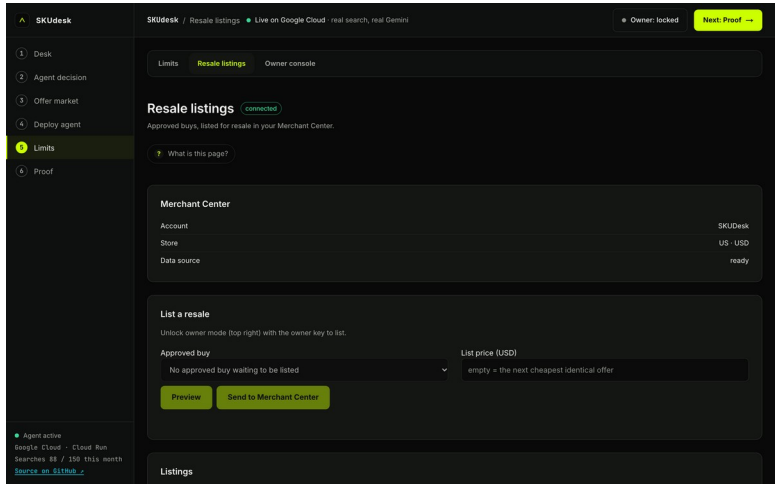












Market page (iPhone 16 Pro, LEGO Ferrari) · Resale listings · Run metrics · Landing · Approvals — live at skudesk.lol

Prototype performance report / benchmarking

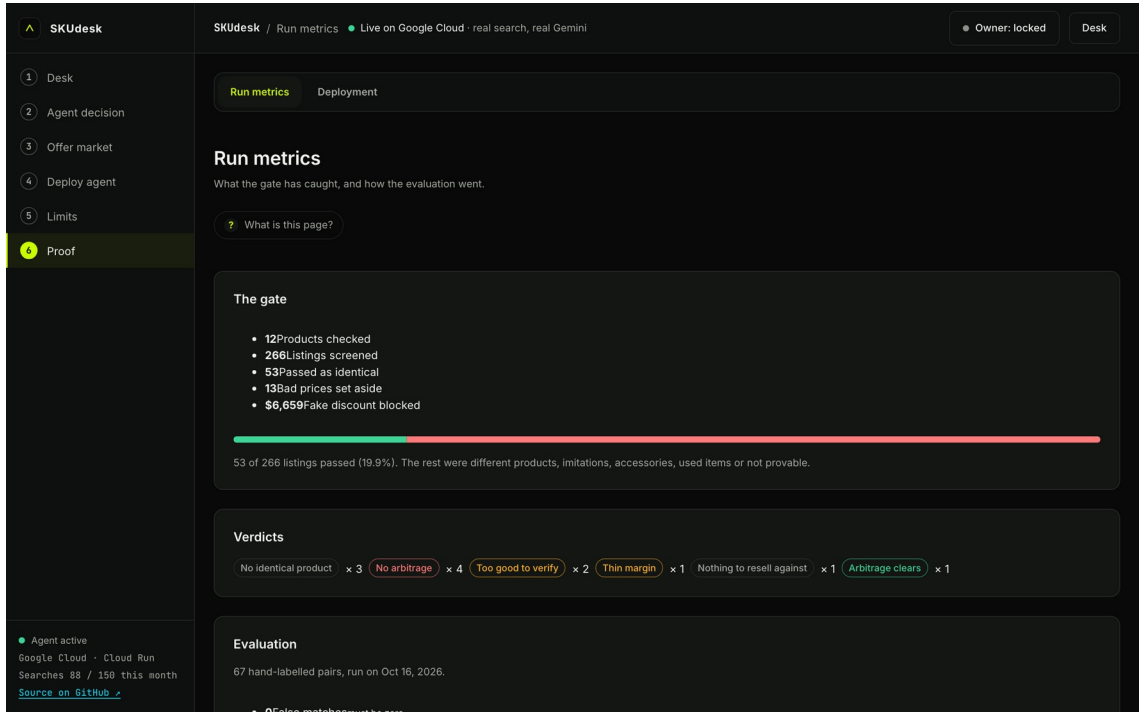
67 hand-labelled product pairs run against the real model: phones, LEGO sets, game consoles, accessories, clones, bundles, regional variants, injected instructions, Thai-language listings.

0 wrong matches (a different product called identical).

0 missed identical products.

64 of 67 exact on the verdict; the other three are “not provable”, which is the safe side.

Live Every check on the site adds to the run metrics: listings screened, passed as identical, bad prices set aside.



Additional details / future development

- Add eBay Browse as a second source (the adapter is built and joins when its key is added).
- Sold-price data for resale pricing, in place of asking prices.
- More categories beyond phones, devices and LEGO, each with its own identity rules and labelled pairs.
- Seller-side agents that bid with their own cost models on the order book.
- Alerts when a watched product opens a gap, with the approval link in the message.

Provide links to your:

- **GitHub public repository** https://github.com/johnnyduo/SKUdesk_
- **Demo video link (3 minutes)** https://youtu.be/DgpX9r0xt9Y
- **Final product link** https://skudesk.lol

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AI Builder Cup

Build in JAPAC.
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Thank you

